

Anomaly Detection: An Introduction

Wigner Summer Camp
Data and Compute Intensive Sciences Research Group

Balázs, Paszkál, Vince, Levente, Bence, Antal
Éva, Hajni

6–10 July 2026



Contents

Why Anomaly Detection?

From Data to Anomalies

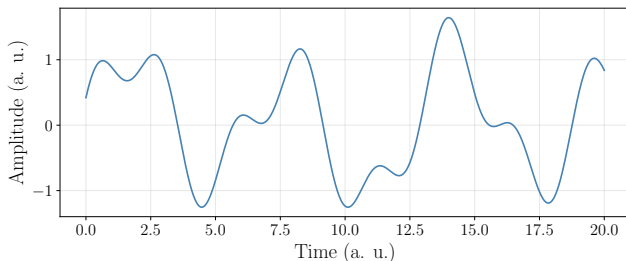
Detecting Anomalies

Why Anomaly Detection Matters

- ▶ Most data is *normal*; the rare exceptions are often what matter most.
- ▶ An **anomaly** can signal a fault, a fraud, a fault-line tremor, or a new discovery.
- ▶ Real-world examples appear everywhere:
 - ▶ A machine vibrating differently just before it breaks.
 - ▶ An unusual transaction hinting at a stolen credit card.
 - ▶ An abnormal heartbeat in a patient's ECG.
 - ▶ A faint, unexpected signal in a physics experiment.
- ▶ These events are **rare**, so we cannot rely on labelled examples of every failure.
- ▶ We must instead learn what *normal* looks like, and flag whatever deviates from it.

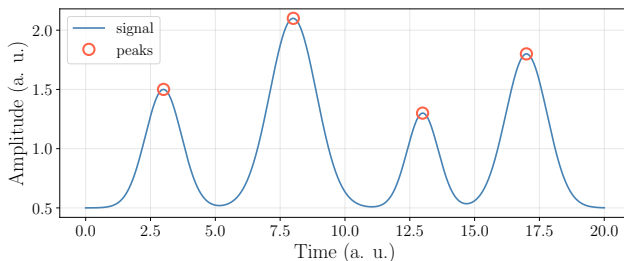
Time Series

- ▶ A **time series** is a quantity measured as a function of time.
- ▶ It is the raw data we work with throughout this camp.
- ▶ Our goal is to find the interesting moments hidden inside it.



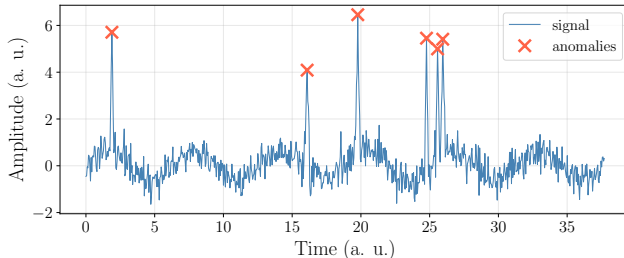
Peaks

- ▶ A **peak** is a sample that is larger than its neighbours.
- ▶ Peaks often mark the events we care about.
- ▶ Detecting them reliably is a core building block of our pipeline.



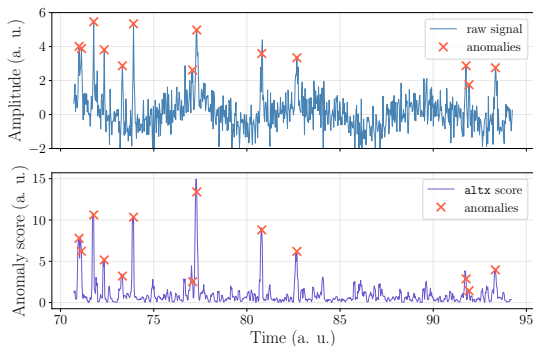
Anomalies

- ▶ An **anomaly** is a sample that breaks the normal behaviour of the signal.
- ▶ Here the normal behaviour is a noisy sine, and the spikes are the anomalies.
- ▶ Finding these rare, unexpected events is the heart of anomaly detection.



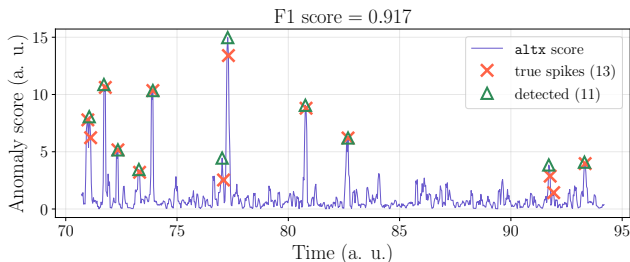
Transforming the Data with `altx`

- ▶ Anomalies can be hard to spot directly in a noisy signal.
- ▶ The `altx` method learns the normal dynamics and turns the signal into an *anomaly score*.
- ▶ This score stays small for normal data and rises sharply at the anomalies.



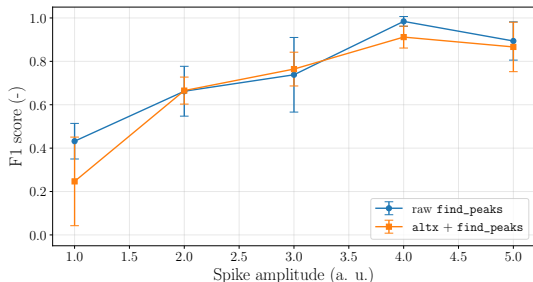
Detecting the Anomalies

- ▶ On the transformed score, the anomalies become clear peaks.
- ▶ We then run peak detection to locate them automatically.
- ▶ The detected peaks match the true anomalies closely.



Does the Transformation Help?

- ▶ We measure detection quality with the **F1 score**, where higher is better.
- ▶ As the anomalies grow stronger, both approaches improve.
- ▶ When anomalies are weak, the **altx** transformation gives a clear advantage.



What Comes Next

Over the coming days we will build this pipeline step by step:

- ▶ The mathematics behind the data and the metrics.
- ▶ Classical signal processing and finding peaks.
- ▶ The `altx` method for transforming time series.
- ▶ Putting it all together into a working anomaly detector.